

MULTIPARAMETRIC ULTRASOUND IN NEOADJUVANT-TREATED MALE BREAST CANCER

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BACKGROUND

Male breast cancer is an uncommon malignancy with limited evidence guiding imaging follow-up during NAC. Multiparametric ultrasound may offer a practical, non-invasive method for assessing tumor response through morphologic, vascular, and tissue stiffness changes. The utility of combined B-mode ultrasound, Doppler, and strain elastography for monitoring NAC response in male breast cancer has rarely been reported.

OBJECTIVES

To describe the role of multiparametric breast ultrasound in monitoring response to neoadjuvant chemotherapy in male breast cancer, with emphasis on changes in tumor size, axillary lymph nodes, vascularity, and tissue stiffness.

CASE PRESENTATION

A 53-year-old man presented with a progressively enlarging painless left breast mass, associated with nipple retraction and intermittent bloody nipple discharge. Core needle biopsy revealed grade III invasive carcinoma of no special type (NST) with lymphovascular invasion and associated high-grade ductal carcinoma in situ (DCIS). The patient underwent neoadjuvant chemotherapy (NAC), followed by modified radical mastectomy. Multiparametric breast ultrasound, including B-mode, Doppler, and strain elastography, was performed before and after NAC to assess treatment response.

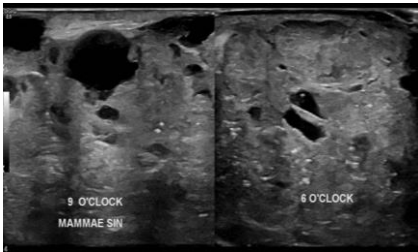


Fig 1. Ultrasound of left breast mass

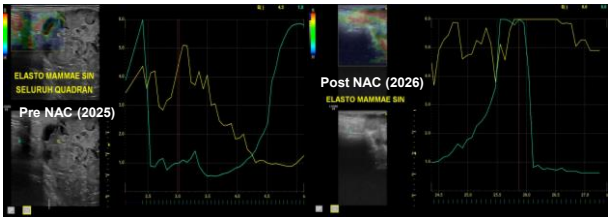
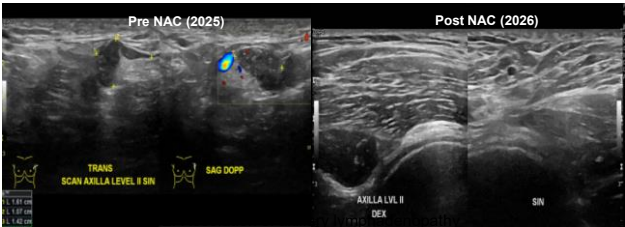


Fig 3. Increased in strain ratio from elastography

Parameter	Before NAC	After NAC
Tumor size	23 × 18 cm	20 × 16 cm
Axillary nodes	Level II lymphadenopathy	Resolved
Strain ratio	4.5	6.0

CONCLUSION

Multiparametric breast ultrasound provides complementary information beyond tumor size in monitoring NAC response

Discordance between dimensional, nodal, and elastographic findings underscores the importance of multiparametric interpretation and pathological correlation for accurate treatment response assessment.

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